	Raffles Institution Physics Department Revision Package 3	Quantities & Measurement	1

Important note: ****denotes questions that may require knowledge from later chapters in order to solve them.***

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/2]

Due to the random nature of radioactive decay, the count rate recorded by a Geiger-Müller tube is subject to statistical fluctuations. When the total count recorded from a source in a given time is *N*, the uncertainty is $\sqrt{N}$.

What is the number of counts taken to obtain a mean count rate precise to 0.1 %?

A 100

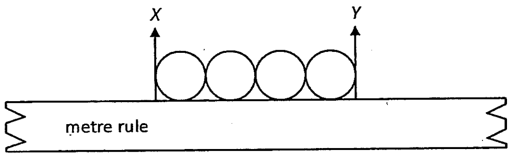
B 1000

C 10000

D 1000000

2. [AJC/H2/Prelims 2018/P1/1]

A student attempts to determine the radius of a steel ball by using a metre rule to measure four similar balls in a row.



The student estimates the position on the scale to be as follows:

$X = (1.00 \pm 0.05)$ cm

$Y = (5.00 \pm 0.05)$ cm

What is the radius of a steel ball together with its associated uncertainty?

A (0.50 ± 0.01) cm B (0.50 ± 0.05) cm C (0.5 ± 0.1) cm D (1.0 ± 0.1) cm

3. [CJC/H2/Prelims 2018/P1/1]

A student measured the mass and linear dimensions of a rectangular block in order to determine its density. The measurements are:

length $= (5.00 \pm 0.01)$ cm ; breadth $= (2.00 \pm 0.01)$ cm ; height $= (1.00 \pm 0.01)$ cm ; mass $= (80.0 \pm 0.2)$ g

What is the uncertainty in the value of the density calculated?

A 0.01 g cm^{-3} B 0.02 g cm^{-3} C 0.1 g cm^{-3} D 0.2 g cm^{-3}

4. [CJC/H2/Prelims 2018/P1/2]

Express the *volt* in SI base units.

A kg m^2 s^{-3} A^{-1} B kg m s^{-2} A^{-1} C J C^{-1} D V

5. [DHS/H2/Prelims 2018/P1/1]

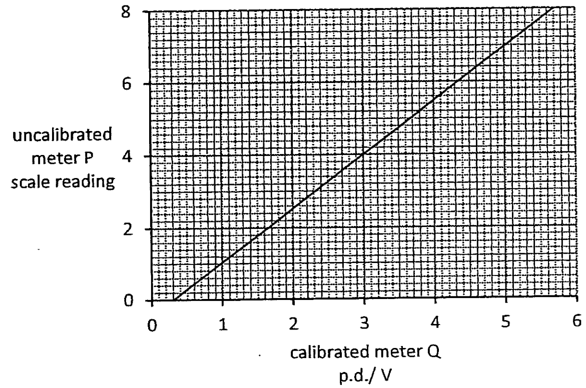
Determine the angle between two equal forces *F* when their resultant force is also equal to *F*.

A 45° B 60° C 120° D 135°

6. [DHS/H2/Prelims 2018/P1/2]

An un-calibrated analogue voltmeter P is connected in parallel with another voltmeter Q which is known to be accurately calibrated. For a range of values of potential difference (p.d.), readings are taken from the two meters.

The diagram shows the calibration graph obtained. The graph shows that meter P has a zero error. This meter is now adjusted to remove this zero error. When the meter is re-calibrated, the gradient of the calibration graph is found to be unchanged.



What is the new scale reading on meter P when it is used to measure a p.d. of 5.0 V?

A 6.6 B 6.7 C 7.2 D 7.4

7. [EJC/H2/Prelims 2018/P1/1]

Measurements are subject to both systematic error and random error.

Which measurements have high accuracy and low precision?

A High random error and high systematic error

B High random error and low systematic error

C Low random error and high systematic error

D Low random error and low systematic error

8. **[HCI/H2/Prelims 2018/P1/1]**

Which of the following statements is correct?

A Density is mass per cubic metre.

B Potential difference is energy per unit current.

C Speed is distance travelled per second.

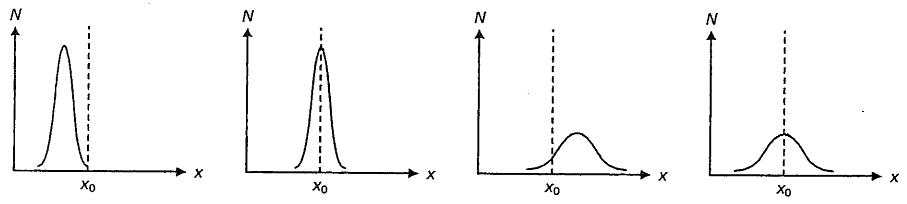
D Pressure is force per unit area.

9. **[IJC/H2/Prelims 2018/P1/1]**

In an experiment, a quantity x is measured many times.

Suppose N is the number of times of measurement which giving a value x and x_0 is the true value of the quantity.

Which of the following graphs best represents measurements of x that are precise but not accurate?



A B C D

10. **[IJC/H2/Prelims 2018/P1/2 Modified]**

Which of the following shows the SI base units for capacitance C?

A $A^2\ s^4\ m^{-2}\ kg^{-1}$ B $s^2\ m^{-2}\ kg^{-1}$ C $A^2\ m^{-2}\ kg^{-1}$ D $C^2\ s^2\ m^{-2}\ kg^{-1}$

11. **[JJC/H2/Prelims 2018/P1/1]**

The speedometer in a car consists of a pointer which rotates. The pointer is situated several millimetres from a calibrated scale.

What could cause a random error in the driver's reading of the car's speed?

A The car's speed is affected by the wind direction.

B The speedometer does not read zero when the car is at rest.

C The speedometer reads 5% higher than the car's actual speed.

D The driver's eye is not always in the same position in relation to the pointer.

12. **[MI/H2/Prelims 2018/P1/1]**

Which of the following pair of physical quantities includes a scalar quantity and a vector quantity?

A work, magnetic flux density

B voltage, frequency

C resistance, charge

D tension, electric field

13. **[MI/H2/Prelims 2018/P1/2]**

The acceleration of free fall g was determined during an experiment by a student. The table below shows the recorded measurements.

g/ m s^{-2}				
9.25	9.23	9.25	9.24	9.23

Which of the following gives a suitable description of the readings?

A The readings were accurate with large systematic error.

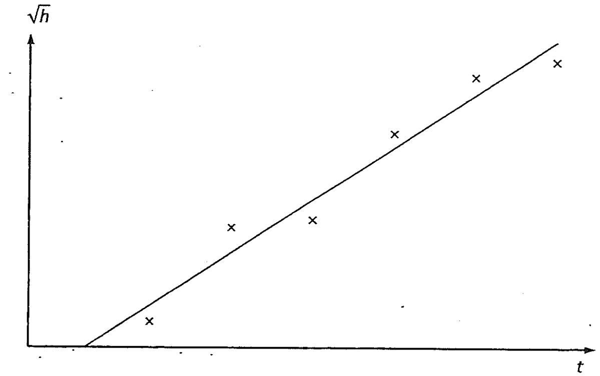
B The readings were accurate with large random error.

C The readings were precise with large systematic error.

D The readings were precise with large random error.

14. **[MJC/H2/Prelims 2018/P1/1 Modified]**

A student measures the time t for a ball to fall from rest through a vertical distance h. The student plots his results and best-fit line in the graph shown.



Which of the following statement is true?

A Data contains systematic error as the data points are scattered about the line.

B Data contains systematic error as the line does not pass through the origin.

C Data contains random error as there are equal number of data points on both sides of the line.

D Data contains no random error as the data points do not deviate significantly from the line.

15. **[MJC/H2/Prelims 2018/P1/2]**

The experimental measurement of the heat capacity of a solid as a function of temperature T is found to fit the following expression

$$C = \alpha T^3 + \beta T$$

What are the possible base units of α and β?

	units of α	units of β
A	kg m^2 s^{-1} K^{-4}	kg m^2 s^{-1} K^{-1}
B	kg^2 m s^{-2} K^{-3}	kg^2 m s^{-2} K^{-2}
C	kg m^2 s^{-2} K^{-4}	kg m^2 s^{-2} K^{-2}
D	kg^2 m s^{-2} K^{-3}	kg^2 m s^{-2} K^{-1}

16. **[NJC/H2/Prelims 2018/P1/1]**

The diameter of a wire, known to be 0.27 mm at room temperature, is measured with an instrument that gives readings to 0.001 mm.

Readings are taken, at room temperature, at three different points along the wire. Two perpendicular values are taken at each point.

The six readings obtained, in mm, are 0.247, 0.247, 0.248, 0.248, 0.249 and 0.247.

Which statement is true?

A The readings are accurate since the spread of the values is within 0.002 mm.

B The readings are precise since all the values are recorded to the third decimal place.

C The readings are inaccurate since the readings are consistently less than the actual value.

D The readings are not precise since the readings are consistently less than the actual value.

17. **[NJC/H2/Prelims 2018/P1/2]**

A radio antenna of length L emits electromagnetic waves of wavelength λ and power P when an alternating current I flows through it. These quantities are related by the expression

$$P = kI^2L^2\lambda^{-2}$$

where k is a constant.

What is the S.I. base unit of the constant k?

A kg m^2 s^{-3} A^{-1} **B** kg m^2 s^{-3} A^{-2} **C** kg m^2 s^{-3} **D** no unit

18. **[NYJC/H2/Prelims 2018/P1/1]**

Which of the following statements is not always true?

A The total mass of two 1 kg masses is 2 kg.

B The total charge of a +1 C charge and a –1 C charge is zero.

C The resultant of two 1 N forces is 2 N.

D The total energy of 1 J of kinetic energy and 1 J of potential energy is 2 J.

19. **[NYJC/H2/Prelims 2018/P1/2]**

The following are the results obtained by two students P and Q in the determination of the mass of an object.

Student P	1.05 kg	0.96 kg	1.01 kg	0.97 kg
Student Q	1.08 kg	1.07 kg	1.08 kg	1.06 kg

Which of the following most appropriately describes the results obtained by students P and Q?

	more accurate	more precise
A	Q	*cannot be determined*
B	P	Q
C	Q	P
D	*cannot be determined*	Q

20. **[PJC/H2/Prelims 2018/P1/1]**

A resistor is marked as having a resistance value R of $4.7\ \Omega \pm 2\ \%$. The current in the resistor is measured as (2.50 ± 0.05) mA.

Given that the power P dissipated in the resistor is $P = I^2R$, what is the percentage uncertainty in the calculated value of P?

A 2 % **B** 4 % **C** 6 % **D** 8 %

21. **[RVHS/H2/Prelims 2018/P1/1]**

The speed v of a liquid leaving a tube depends on the change in pressure ΔP and the density ρ of the liquid.

The speed is given by the equation

$$v = k\left(\frac{\Delta P}{\rho}\right)^n$$

where k is a constant that has no units.

What is the value of n?

A $\frac{1}{2}$ **B** 1 **C** $\frac{3}{2}$ **D** 2

22. **[SRJC/H2/Prelims 2018/P1/1 Modified]**

The rate of heat flow, R, can be found using the equation

$$R = kA \cdot \frac{(T_H - T_L)}{L}$$

where k is the thermal conductivity, A is the total cross-sectional area of the conducting surface, and L is the thickness of conducting surface separating the two temperatures T_H (hot) and T_L (cold).

What is the SI base units for k?

A kg m s^{-3} K^{-1} **B** kg m s^{-2} K^{-1} **C** kg m s^{-3} **D** kg s^{-2} K^{-1}

23. [SRJC/H2/Prelims 2018/P1/2 Modified]

With reference to Question 22 above, a student collected the following measurements to determine the rate of heat flow R of the object:

$$L=(0.35\pm0.05)\text{ mm}$$
$$T_H=(32.0\pm0.5)\ ^\circ\text{C}$$
$$T_L=(4.0\pm0.5)\ ^\circ\text{C}$$

What is the fractional uncertainty in R?

A 0.11 B 0.14 C 0.16 D 0.18

24. [SRJC/H2/Prelims 2018/P1/3]

A value for the acceleration of free fall on Earth is given as $(10\pm2)\text{ m s}^{-2}$.

Which statement is correct?

A The value is accurate but not precise.

B The value is both precise and accurate.

C The value is neither precise nor accurate.

D The value is precise but not accurate.

25. [TPJC/H2/Prelims 2018/P1/1]

The relation between the velocity v of waves in the sea with its wavelength λ, the surface tension γ and density ρ of sea water is given by

$$v=k\sqrt{\frac{\gamma}{\lambda\rho}}$$

where k is the constant of proportionality.

If $\gamma=(4.30\pm0.05)\text{ N m}^{-1}$, $\rho=(1450\pm20)\text{ kg m}^{-3}$ and the percentage uncertainty in λ is 5 %, what is the percentage uncertainty in the velocity of the waves?

A 2 % B 3 % C 4 % D 8 %

26. [TPJC/H2/Prelims 2018/P1/2]

A cylindrical tube rolling down a slope of inclination θ moves a distance L in time T. The equation relating these quantities is

$$L\left(3+\frac{a^2}{P}\right)=QT^2\sin\theta$$

where a is the internal radius of the tube and P and Q are constants.

Which of the following gives the correct units for P and Q?

	units of P	units of Q
A	m	m s^{-2}
B	m^2	m s^{-2}
C	m^2	$\text{m}^3\text{ s}^{-2}$
D	m^2	$\text{m}^2\text{ s}^{-2}$

27. [VJC/H2/Prelims 2018/P1/1]

To find the resistivity of a semiconductor, a student makes the following measurements of a cylindrical rod of the material.

$$\text{length}=(25\pm1)\text{ mm}$$
$$\text{diameter}=(5.0\pm0.1)\text{ mm}$$
$$\text{resistance}=(68\pm1)\ \Omega$$

He calculates the resistivity to be $5.341\times10^{-2}\ \Omega\text{m}$.

How should the uncertainty be included in his statement of the resistivity?

A $(5.34\pm0.07)\times10^{-2}\ \Omega\text{ m}$

B $(5.34\pm0.09)\times10^{-2}\ \Omega\text{ m}$

C $(5.3\pm0.4)\times10^{-2}\ \Omega\text{ m}$

D $(5.3\pm0.5)\times10^{-2}\ \Omega\text{ m}$

28. [YJC/H2/Prelims 2018/P1/1]

A student counts the oscillations of a simple pendulum but mistakenly starts by counting "1" instead of "0" when the bob is released. He finishes at a count of "10".

What is the percentage error in the student's calculation of the period, due to this mistake?

A 10 % low B 10 % high C 11 % low D 11 % high

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P2/1]

a. State two differences between random and systematic errors. [2]

b. Fig. 1.1 shows a simple pendulum consisting of a mass m attached to a light inextensible string of length L. The pendulum is secured to a fixed point and made to undergo oscillations when displaced sideways by a small angle θ.

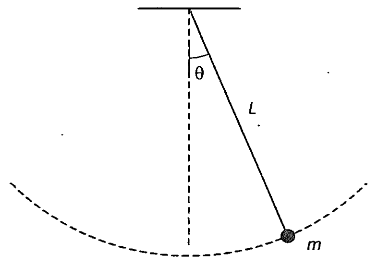


Fig. 1.1 (not to scale)

The period of oscillation T can be found by the following equation:

$$T^2=4\pi^2\cdot\frac{L}{g}$$

where g is the acceleration of free fall.

i. Given $L = 50.0 \pm 0.2$ cm; $g = 9.8 \pm 0.1$ m s^{-2}, find T with its associated uncertainty. [4]

ii. When determining the period of oscillation using a stopwatch, it is a good practice to take a large number of oscillations such that the total time taken is more than 20 seconds.
Explain why this should be done. [1]

iii. The acceleration of free fall g on a body is caused by the gravitational field of the Earth.
Suggest why there is a variation in the strength of the Earth's gravitational field at different locations on Earth. [1]

[Total: 8]

2. **[DHS/H2/Prelims 2018/P3/1 Modified]**

a. Make estimates of the following quantities.

i. the thickness of a sheet of A4 paper [1]

ii. the mass of a sheet of A4 paper [1]

b. The distance from the Earth to the Sun is 0.15 Tm.
Calculate the time in minutes for light to travel from the Sun to the Earth. [2]

c. The time T for a satellite to orbit the Earth is given by

$$T = \sqrt{\frac{KR^3}{M}}$$

where R is the distance of the satellite from the centre of the Earth, M is the mass of the Earth and K is a constant.

i. Determine the SI base units of K. [2]

ii. Data for a particular satellite are given in Fig. 2.1.

quantity	measurement	uncertainty
T	8.64×10^4 s	0.50%
R	4.23×10^7 m	1.0%
M	6.00×10^{24} kg	2.0%

Fig. 2.1

Express K with its associated uncertainty in SI units. [3]

iii. State the quantity which contributes the largest uncertainty in the value of K. [1]

[Total: 10]

3. **[JJC/H2/Prelims 2018/P3/1]**

In a simple pendulum experiment to determine the acceleration of free fall g, the following equation is used

$$T = 2\pi\sqrt{\frac{L}{g}}$$

The following measurements were obtained using a metre rule and stopwatch respectively.

length of pendulum $L = (98.0 \pm 0.1)$ cm

average time of 10 oscillations $t = (19.8 \pm 0.2)$ s

a. i. The calculated acceleration of free fall g is 9.869 m s^{-2}.
Determine the acceleration of free fall with its associated uncertainty. [3]

ii. Using a different set of apparatus, another student obtained g to be (11.15 ± 0.02) m s^{-2}.
Compare the accuracy and precision of the two sets of readings. [1]

b. Tempered glass screen protector is made of many atoms.
Estimate the number of atoms in a 0.5 mm thickness tempered glass screen protector for a mobile phone.
Show your working and reasoning clearly. [4]

[Total: 8]

4. **[MI/H2/Prelims 2018/P2/1]**

a. A speed boat has an initial velocity of 12 m s^{-1} south. The speed boat then changes its course to a final velocity of 16 m s^{-1} east. Determine the magnitude of the change in velocity, Δv, of the speed boat, and the angle between vector Δv and the north direction. [2]

b. A student uses a pair of vernier calliper to take several measurements of the diameter of a rubber ball. He applies different pressures when closing the gap of the vernier calliper.
State and explain whether this will introduce a systematic error or random error into the readings. [2]

c. The energy per unit time, P radiated by an object with a surface area A at thermodynamic temperature T is given by

$$P = e\sigma AT^4$$

where e is the emissivity of the surface and σ is the Stefan-Boltzmann constant. [2]

i. Given that the SI unit for σ is W m^{-2} K^{-4}, determine the base units of emissivity, e. [2]

ii. In an experiment to determine emissivity e of a circular surface area of diameter d of an object, the following measurements are taken:

$P = (3.0 \pm 0.2)$ W

$d = (5.0 \pm 0.1)$ cm

$T = (500 \pm 1)$K

$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

Determine the value of emissivity, e of the surface and express it with its associated uncertainty. [4]

[Total: 10]

5. [MJC/H2/Prelims 2018/P3/2]

In an experiment to determine the specific heat capacity of a liquid, a student heated a fixed mass of the liquid for a fixed duration of time, using an electric heater. The student repeated the experiment three times to find the rise in temperature of the liquid. The following measurements were obtained:

Mass of liquid, m	309 ± 3 g
Voltage applied across heater, V	11.8 ± 0.3 V
Current flow in the heater, I	4.125 ± 0.002 A
Time taken, t	200.0 ± 0.5 s

The rise in temperature θ was recorded for each attempt:

Attempt:	1st	2nd	3rd
θ / K	10.2	9.7	10.5

a. Estimate the uncertainty in θ. [1]

b. Calculate the specific heat capacity c of the liquid. [2]

c. Calculate the uncertainty in specific heat capacity c of the liquid and express the specific heat capacity c together with its uncertainty. [3]

d. State an assumption made in your calculation of the specific heat capacity c of the liquid. [1]

[Total: 7]

6. [NYJC/H2/Prelims 2018/P3/1 Modified]

a. The SI unit for capacitance is the farad (F). Express F in SI base units. [3]

b. A capacitor of capacitance 50 μF is charged using a cell in the circuit shown in Fig. 6.1.

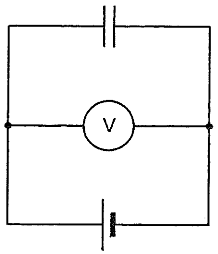


Fig. 6.1

i. The voltmeter reads 2.20 V. Calculate the charge stored in the capacitor to 3 significant figures. [1]

ii. The uncertainty of the voltmeter reading is 0.05 V, and the capacitance of the capacitor has an uncertainty of 10%.

Express the charge stored in the capacitor in the form of $(Q \pm \Delta Q)$. [3]

[Total: 7]

7. [RVHS/H2/Prelims 2018/P2/1 Modified]

A student sets up the circuit shown in Fig. 7.1 in order to determine the resistivity of a piece of wire.

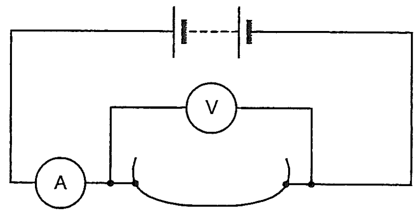


Fig. 7.1

The following readings were obtained:

Reading of voltmeter, V	Reading of ammeter, I	Length of wire, L	Diameter of wire, d
1.50 ± 0.01 V	0.82 ± 0.01 A	63.2 ± 0.1 cm	0.48 ± 0.02 mm

a. Show that the resistivity is given by $\rho = \dfrac{\pi d^2 V}{4IL}$. [1]

b. Hence or otherwise, calculate the resistivity of the wire together with its associated uncertainty. [4]

[Total: 5]

8. [VJC/H2/Prelims 2018/P3/1 Part]

a. Group the following terms into pairs, so that in each pair, a quantity is given followed by its corresponding unit.

power, volt, watt, magnetic flux, magnetic flux density, potential difference, tesla, weber. [2]

b. Explain why it is technically incorrect to define speed as distance travelled per second. Include in your answer the correct statement defining speed. [2]

[Total: 4]

9. [YJC/H2/Prelims 2018/P2/1]

Archimedes' number, A_r, is dimensionless (unitless) and is used in the study of objects in fluids. The number is given by the following expression

$$A_r = \frac{gL^3 \rho_l (\rho - \rho_l)}{\mu^2}$$

whereby g is the acceleration due to gravity, L is the characteristic length of the object, ρ_l is the density of the fluid, ρ is the density of the object and μ is the dynamic viscosity of the fluid.

Determine the units of μ. [3]

[Total: 3]